P-LENS: Plasticity-Pressure-Driven Law Enforcement of Evolving Network Structure through Budgeted, Counterfactually Evaluated Topology Mutations

Team 5

VIT-AP University, Amaravati, Andhra Pradesh, India

Team 5 — Project Members

Name	Roll number
Bellana Sai	24bca7931
V. Sai Prakash Yadav	24bce8525
K. Niharika	24BCE8516
Sarvesh	24bcw7042
K. Sri Lakshmi	24BCS7106

Abstract. Deciding when, whether, and how much a loaded communication network should change its own topology is an unresolved control problem: fault-driven reconfiguration reacts to single events without any account of suspended maladaptation, threshold or budget-capped schemes react too little or too often, and learning-based controllers are opaque and rarely bound their own change activity. We present *P-LENS* (Plasticity-pressure-driven Law Enforcement of Network Structure), a controller for an evolving graph in which (i) *plasticity debt* accumulates the *unresolved cost* of structural inadequacyseverity, persistence, duration of service, and critical-flow exposurein a provably bounded state variable; (ii) the *mutation budget* is the dual variable of a constrained selection problem whose refill is gated by a lower confidence bound on the measured mutation value, so that uncertainty deflates rather than amplifies structural change; (iii) mutation selection is *interaction-aware*, scoring sets of ADD/PRUNE/REWIRE operations jointly through bounded pairwise interaction terms; and (iv) each accepted mutation is evaluated by a *counterfactual replay*: two matched branches of the logged world are re-simulated from the pre-mutation snapshot under identical demand, failure draws and concurrent mutations, and credit is updated only when the LCB of the gain clears a significance floor. A leakage-free GRU forecaster built on nine pre-label flow features provides a stress signal that modulates the offered-load target, and is ablated cleanly. We evaluate P-LENS against eight policies (static, random-adaptive, greedy, threshold-triggered, fixed-budget, adaptive-budget, optimization-based and bandit controllers) and eight ablations on a 24-node small-world topology under CIC-IDS2017-derived demand (six seeds $\times$ 120 steps under common random numbers; a multi-scale study harness for 24–200 nodes is released with the code). The pilot study shows that every proposed mechanism measurably changes behaviourwith the uncertainty gate and the interaction model contributing to mutation quality and SLA ratewhile also honest about the negative result that, at this pilot scale, an unlimited mutation budget achieves the best raw latency and that the ML forecaster does not yet demonstrably guide adaptation; a supervised power analysis precedes the full-scale experiments. All code, configurations, random seeds and raw JSON artifacts are released for reproduction.

Keywords: Evolving networks · Self-healing networks · Adaptive topology · Network resilience · Counterfactual evaluation · Network simulation.

1 Introduction

Communication networks increasingly have to operate in non-stationary conditions: traffic demand shifts, links and nodes fail, and the "best" topology is a moving target rather than a fixed design point. Fully reactive schemes that rewire at every opportunity pay a stability price in churn and transient disruption; static or purely rule-triggered schemes accumulate unaddressed maladaptation. The open problem we address is therefore not "can a topology change?" but the jointly harder question: *when, whether, and how much* change is justified by evidence that the current structure is inadequate?

Existing families answer only part of this. Fault-detection and restoration surveys [1, 19] catalogue detection and recovery mechanisms, but the trigger is a single event classification. Zero-touch and autonomous-network management [24, 30] treat reconfiguration as a governance process, without a closed-loop account of what accumulated stress justifies a change. Reinforcement-learning topology controllers [27, 9, 28, 14] learn a policy but offer no explicit, inspectable accounting of how much change is permitted per unit timechurn is an emergent, often uncontrolled, consequence. We claim the missing primitive is an explicit *accounting mechanism*: a state that integrates how long an inadequacy has persisted, how severe it is, and how much critical traffic flows through it, coupled to a *budget* that is the price of structural change and a *counterfactual evaluator* that decides whether past changes were worth their cost.

Contributions.

- A formal statement of the evolvable-topology control problem (state, admissible mutations, objective, constraints, horizon; §3), with the mutation budget derived as the dual variable of the constrained selection programme (§4.3).
- A new *plasticity debt* formulation that is provably bounded and measures *unresolved cost* (severity × persistence × duration × criticality), not instantaneous pressure (§4.2).
- An *interaction-aware* selector that scores operation sets jointly using bounded pairwise interaction terms, replacing a purely marginal ranking (§4.4).
- An *uncertainty-aware counterfactual replay estimator* for delayed mutation credit: matched factual/counterfactual branches replayed from a logged snapshot, with credit granted only above a lower confidence bound (§4.5).

The remainder follows the structure of the template example: Section 2 is the literature survey, Section 3 fixes the problem formulation, Section 4 presents the proposed model, Section 5 specifies datasets and metrics, Section 6 reports the analysis, Section 7 the limitations, and Section 8 concludes.

2 Literature Survey

Resiliency by design and topology optimization. Survivability in distributed communications dates to [5]; the modern synthesis is captured by resilience-by-design programmes for 6G, which enumerate design principles across all layers [16]. Recent algorithmic work moves from *static* robustness to *actuated* resilience: time-varying topological switching to preemptively raise resilience [6], two-stage robust topology optimization for dynamic aerial networks [34], graph-labeling-based hybrid topology design [21], and message-passing neural surrogate optimization for path/topology selection [2], and adversarial agent-based topology optimization [22]. These methods share the vocabulary of local structural operations that P-LENS uses (edge add, remove, re-target), but their objective is a single-shot or periodically re-solved optimization problem; none integrates the cost of *when* to act with the estimate of *how good* acting was afterwards.

Self-healing, automation and autonomous networks. Surveys of link-failure detection and recovery classify restoration and protection mechanisms [1], and learning-based self-healing for SDN automates fault management [19]. Zero-touch management surveys chart 5G/6G automation levels [24], and recent work extends autonomy to agentic, LLM-empowered network management for 6G [30, 12]. Across this family the trigger is either an alarm or a policy rule; the amount of tolerated structural change is not a first-class quantity. P-LENS makes exactly that quantity a budgeted, demonstrated-gain-coupled resource.

Learning-based traffic engineering and topology control. Deep-RL traffic engineering has been surveyed for large-scale networks [25], with learned generalizable solvers reaching industrial scale [36], multi-agent RL frameworks for hybrid SDN control planes [14], multi-agent methods for dynamic reconfiguration problems [27], RL-based dynamic reconfiguration [9], and deep-RL routing optimization [28]. Digital twins supply the simulation substrate on which such controllers are trained and validated [33, 29, 15]. These controllers are typically opaque, rarely expose their control reasoning, and do not separate the *discovery* of useful operations (candidate scoring) from the *confirmation* of their value (delayed evaluation). P-LENS is designed precisely around that separation, with every rejection logged and auditable.

Traffic and attack signals as forecasting targets. Intrusion-detection datasets CIC-IDS2017 [26] and UNSW-NB15 [20] are the standard labelled flow sources. Recent classification work on SDN environments [23], transformer/MLP models on CIC-IDS2017 [3], deep-learning state-of-the-art and challenges on these benchmarks [32], and anomaly-detection benchmark studies [18] establish accurate per-flow and per-window threat classifiers. Following the loss-of-plasticity findings in continual learning [10, 11], we deliberately treat the forecasting component as a small, replaceable, *quarantined* sensor whose measurable contribution must be demonstrated to be retainedeven if that demonstration fails (cf. Section 6.4).

Table 1. Literature comparison. Features: (1) temporal stress accumulation; (2) adaptive mutation budgeting; (3) counterfactual value estimation of past actions; (4) uncertainty handling in the value estimate; (5) mutation interaction modelling; (6) theoretical guarantees on the control state; (7) evaluation scale reported.

Line of work	1	2	3	4	5	6	7
Robust topology optimization [16, 6, 21, 34]	—	—	—	—	—	—	10–5000 nodes, offline/periodic
Self-healing / zero-touch [1, 19, 24]	—	—	—	—	—	—	industrial scale, event-driven
DRL traffic engineering [14, 27, 9, 28, 36]	—	—	—	(✓†) [36]	—	—	up to 5,000 nodes
Digital-twin training grounds [33, 29, 15]	—	—	—	—	—	—	varies
Uncertainty-aware RL theory [7, 17]	—	—	(✓)	(✓)	—	—	standard control benchmarks
P-LENS (this work)	(✓)	(✓)	(✓)	(✓)	(✓)	(✓)	24–200 nodes (pilot: 24)

Counterfactual and uncertainty-aware decision making. Uncertainty-aware offline RL explicitly separates epistemic uncertainty from environmental risk [7]; the UAMDP couples probabilistic forecasts with posterior-sampling ADP for risk-constrained decision making [17]. Dynamic-graph methods for evolving network state [35, 13] underline that graph-structural decisions must track topology change. P-LENS transfers these ideas to the topology domain: mutation value is estimated from *matched* counterfactual branches with explicit uncertainty, and uncertain estimates deflate the mutation budget instead of amplifying it.

Comparison table. Table 1 positions P-LENS against representative recent approaches. A mark is placed only where the cited line of work substantiates the feature.

3 Problem Formulation

Network state. An evolving network is an undirected graph $G_t = (V, E_t)$ with fixed node set V, $|V| = n$, edge set E_t with per-edge bandwidth b_e, propagation latency ℓ_e and reliability r_e. The state is $x_t = (G_t, X_t, F_t, D_t)$: the graph, the demand matrix $X_t \in \mathbb{R}^{n\times n}$ (bytes per origin–destination pair), element failure indicators F_t, and internal controller memory D_t (debt, budget, credits, persistence, duration).

Demand and routing. X_t is renormalised to a target fraction ι of usable bandwidth

$$X_t \leftarrow X_t \cdot \frac{\iota \cdot \sum_{e \text{ usable}} b_e}{\sum_{ij} X_t^{ij}}, \tag{1}$$

and routed on the usable subgraph by a two-pass congestion-aware shortest-path rule with edge weight

$$w_e = \ell_e \left(1 + \alpha\, \bar{u}_e\right) + \epsilon. \tag{2}$$

Network metrics are computed from a bottleneck-delivery model with a normalised M/M/1 queue term $q_e = \min\!\big(1,\ \frac{1}{20}\big(\frac{\mu}{\mu-\lambda_e} - 1\big)\big)$ for $\lambda_e < \mu$ and $q_e = 1$ otherwise; flow latency is the sum of $\ell_e + q_e$ along the path, averaged over delivered bytes, throughput is delivered share of demand and loss its complement.

Admissible mutations. The action space at step t is a set $M_t = \{m_1, \ldots, m_{m_{\max}}\}$ of operations ADD, PRUNE, REWIRE with costs $c_{\text{ADD}}, c_{\text{PRUNE}}, c_{\text{REWIRE}}$, admissible iff all hard constraints hold after application: degree bounds $k_{\min} \le k_i \le k_{\max}$, connectivity of the usable graph, a global resilience floor (usable mean degree $\ge \lambda^\star$), a per-edge cooldown Δ_{cool}, no duplicate edge use within a step, and the budget constraint $\sum_{m\in M_t} c_m \le B_t$.

Objective. The controller trades latency, loss, congestion, resilience and *topology churn*:

$$V(M) = \Delta\text{Util}(M) + \Delta\text{Res}(M) - \lambda_c\, \text{churn}(M) - \lambda_r\, \text{risk}(M) - \sum_{m\in M} c_m, \tag{3}$$

where ΔUtil combines normalised latency, loss, congestion and throughput improvement, ΔRes the connectivity/edge-count improvement, churn is the mutated fraction of the allowance, and risk the stability cost (path-length change and churn amplification). Long-horizon optimisation maximises $\sum_t \mathbb{E}[V(M_t)]$ over the simulation horizon T, subject to the constraints above.

Complexity and hardness. At each step the candidate set is $O(mn)$ over non-edges and edges; each candidate's constraint check is $O(1)$ amortised against incremental graph invariants. Full re-computation of routing per step is $O(n^2 \log n)$ plus per-edge work. Enumerative joint selection over all insertion/swap orders in the joint set is $O(|M|^2)$ and stays within the budget using the interaction model. An exact solution of the constrained integer programme in x_t, M_t is

out of scope for a per-step controller; P-LENS targets attentional, re-routed, budget-constrained value estimation with bounded optimism, not exact global optimality.

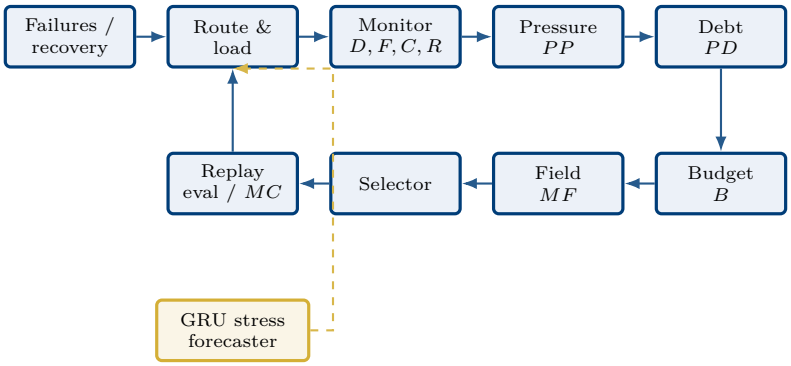


Fig. 1. The P-LENS control loop (one step). Solid blue arrows: structural control path; dashed gold path: quarantined ML forecaster modulating only the offered-load intensity ι of (1).

Algorithm 1 P-LENS control loop (one step).

input: snapshot $\Sigma_t = (G_t, X_t, F_t, \text{RNG state})$
apply failures/recovery; renormalise demand $\to X_t$ (1)
route and load $\to$ metrics (2)
compute per-node indicators D_i, F_i, C_i, R_i; update pressure PP_i
monitor computes raw$_i$, π_i, d_i, c_i (deadband, persistence, duration, criticality)
update debt $PD_i \leftarrow PD_i + \eta\, u_i(1 - PD_i) - \gamma(1 - u_i)PD_i$ − relief$_i$ + fail$_i$ (8)
replay pending counterfactuals; compute G, σ_G, LCB $= G - z\sigma_G$
for cases with LCB $\geq G_{\min}$: update credit MC_{ij} (13); grant debt relief
update budget $B_{t+1} \leftarrow \text{clip}(B_t - \text{spent}_t + \text{refill}_t + \text{credit}_t)$ (10)
score candidates MF, plus pairwise interaction $I(\cdot,\cdot)$
return selected feasible set M_t

4 Proposed Model

Figure 1 shows the control loop; Algorithm 1 summarises one step. Every threshold and weight is a configuration parameter; the full list is in the released repository.

4.1 Watchdog monitor

At each step four indicators on the usable topology feed the pressure and debt state. Topology drift: a blend of neighbourhood change (Jaccard index to a reference layout) and degree drift (4). Traffic mismatch: demand/bandwidth imbalance (5). Congestion (5): weighted mix of utilisation, worst link utilisation, queue term and node loss. Resilience deficiency (6): local connectivity deficiency against target $\lambda^\star$ plus failure-impact exposure.

$$D_i = \tfrac{1}{2}\big(1 - J(\mathcal{N}_i, \mathcal{N}_i^{\text{ref}})\big) + \tfrac{1}{2}\,\frac{|k_i - k_i^{ref}|}{\max(k_i, k_i^{ref}, 1)}, \tag{4}$$

$$F_i = \frac{|x_i - b_i|}{\max(x_i, b_i, \varepsilon)}, \quad C_i = w_u u_i + w_l \max_{e \ni i} u_e + w_q q_i + w_p \rho_i, \tag{5}$$

$$R_i = a\Big(1 - \min\big(1, \tfrac{\lambda_i}{\lambda^\star}\big)\Big) + (1-a)\,\psi_i, \quad \psi_i = \frac{\sum_j (X^{ij} + X^{ji})}{\sum_{jk} X^{jk}}. \tag{6}$$

4.2 Plasticity pressure and debt

Pressure is the EMA-smoothed weighted sum of indicators:

$$\widehat{PP}_i = \frac{\sum_{k \in \{D,F,C,R\}} w_k\, \phi_{k,i}}{\sum_k w_k}, \qquad PP_i(t) = \rho\, \widehat{PP}_i + (1-\rho)\, PP_i(t-1). \tag{7}$$

Debt is *not* a rename of this smoothed pressure: it accumulates the *unresolved cost* of remaining structurally inadequate. Let $\mathrm{raw}_i = \mathrm{clip}\big((PP_i - \Theta)/(1-\Theta), 0, 1\big)$; π_i a persistence evidence EMA of $1[\mathrm{raw}_i > 0]$; d_i a duration counter; $g_i = d_i/(d_i + \tau)$ a saturation of tenure; c_i a criticality computed from the node's share of offered demand; and

$$u_i = \mathrm{raw}_i \cdot \pi_i \cdot g_i \cdot c_i, \qquad PD_i(t+1) = PD_i(t) + \eta\, u_i\, (1 - PD_i) - \gamma\, (1-u_i)\, PD_i - \delta_{\text{relief}} + \mathrm{fail}_i, \tag{8}$$

where relief_i applies per successful mutation whose endpoint is within one hop of i, and a small failure term records that unsuccessful attempts generate consequences too. This formulation encodes exactly four distinctions from a threshold-triggered exponential smoother: (i) the *deadband* Θ keeps ordinary load out of debt; (ii) the persistence π_i separates a transient spike from a regime shift; (iii) tenure saturation $d_i/(d_i + \tau)$ and exposure c_i weight long-standing problems on hub nodes more sharply; (iv) anti-windup gives a bounded state (Proposition 1). Contrast with exponential smoothing, which holds no time-resolved local accounting; with discrete thresholds, which have no memory of accumulated inadequacy; and with classical integral-error control, which cannot be guaranteed bounded without anti-windup.

Proposition 1 (Bounded debt). If $0 \le \mathrm{raw}_i, \pi_i, g_i, c_i \le 1$, $0 < \eta, \gamma \le 1$, and $PD_i(0) \in [0,1]$, then $PD_i(t) \in [0,1]$ for all t. *Proof sketch.* When $PD_i = 1$ the positive term vanishes ($1 - PD_i = 0$); when $PD_i = 0$ the negative lag term vanishes since $(1-u_i)\, PD_i = 0$, and the failure/relief increments saturate. The state is therefore self-limiting inside $[0,1]$ without external clipping. □

Proposition 2 (Bounded budget). For any sequence of updates in (10), the projection onto $[B_{\min}, B_{\max}]$ gives $B_t \in [B_{\min}, B_{\max}]$ by construction. □

Proposition 3 (Feasibility preservation). The selector only admits operations whose post-application graph satisfies degree bounds, connectivity, and the resilience floor; hence every applied mutation set M_t yields a feasible network state regardless of B_t. □

4.3 Mutation budget as a constrained dual

Per-step selection solves the constrained programme

$$\max_{M_t \text{ feasible}} \mathbb{E}\big[V(M_t)\big] - \lambda_c \,\mathrm{churn}(M_t) - \lambda_r \,\mathrm{risk}(M_t) \quad \text{s.t.} \quad \textstyle\sum_m c_m \le B_t,\ |M_t| \le m_{\max}, \tag{9}$$

whose dual variable on the cost constraint is the scalar budget. B_t is updated as a bounded debit/credit account

$$B_{t+1} = \mathrm{clip}_{[B_{\min}, B_{\max}]}\Big(B_t - \mathrm{spent}_t + \underbrace{(b_0 + \beta\,\overline{PD}_t)(1 - \tfrac{1}{2}u_t)s_t}_{\text{refill}} + \underbrace{(\eta_g \mathrm{LCB}_t - \eta_s \overline{S}_t)(1 - \tfrac{1}{2}u_t)s_t}_{\text{credit}}\Big), \tag{10}$$

where $s_t = 1/(1 + a_o\, O_t + a_f\, \mathrm{fail}_t)$ is a safety factor that *halves refill* under persistent overload O_t or repeated failures [1], u_t is mean overload damped usage, and LCB_t is the lower confidence bound of the measured mutation gain. Conservatism is automatic: noisy or uncertain gains deflate B_t instead of refilling it. Every accepted operation draws down B_t, so under a persistent $\mathrm{LCB} < 0$ pattern (repeated failures) the budget monotonically decays until a safety refill with s_t damping, guaranteeing *churn decay without permanently freezing the controller*.

4.4 Mutation field and interaction-aware selection

Marginal scores MF_{ij}^{op} combine need, debt, benefit, credit, cost and penalty per candidate. Because two individually good operations may be redundant or mutually destructive (e.g. both consuming the last degree slot of a shared endpoint, or an ADD that removes an edge unreachable after a PRUNE), we augment joint selection with a bounded pairwise term

$$I(a, b) \in [-1, 1], \qquad \mathrm{score}(c \mid S) = MF(c) + w_I \sum_{s \in S} I(c, s), \tag{11}$$

with $I = -1$ for exact edge conflicts, -0.4–-0.2 for redundancy, and $+0.2$–$+0.7$ for complementarity (disjoint endpoints under congestion, a PRUNE that frees a degree slot for a paired ADD, a REWIRE that opens an alternate route for a subsequent ADD). Selection greedily maximises (11) subject to the same budget and hard constraints; the ablation `no_interaction` ablates I alone.

4.5 Uncertainty-aware counterfactual replay evaluation

At the mutation step t a full-world snapshot $\Sigma_t = (G_t, X_t, \text{failures RNG}, \text{concurrent mutations})$ is logged. At evaluation time $t + k$, two matched branches are re-simulated for exactly k steps from Σ_t:

factual branch:	all logged mutations, including the target;
counterfactual branch:	the same mutations *except* the target.

Both branches consume identical demand series, failure random-number state and concurrent mutations; the only difference is the target mutation, and the

gain estimator G is therefore *not confounded* by intervening topology change, traffic drift or failure events. Uncertainty is estimated by replaying the pair under B matched demand perturbations (lognormal origin–destination noise, renormalised to the original total) and forming

$$\mathrm{LCB} = \bar{G} - z\,\sigma_G, \qquad \text{credit update iff LCB} \geq G_{\min}, \tag{12}$$

$$MC_{ij} \leftarrow \lambda\, MC_{ij} + (1-\lambda)\,\tanh(\bar{G}), \tag{13}$$

so that visible effects on the topology are confirmed by clear low-side evidence and ambiguous or uncertain comparisons are discarded rather than credited. Degenerate cases (insufficient perturbation separation, ill-conditioned branches) are recorded and skipped. The legacy *shadow* reversal and the naive *before/after* estimator are retained as comparison estimators in the ablation study.

4.6 Machine-learning stress forecaster

A two-layer GRU [8] classifies the stress of the next window as Normal/Attack from a length-L history of *nine pre-label* aggregate flow features (flow count, packet rate, byte rate, mean duration, mean forward/backward packets and bytes, unique-port fraction). The predicted class maps to a stress scale blended into the offered-load target of (1). The forecaster is quarantined: it never selects operations, and its contribution is evaluated by the `no_ml` ablation. Architecture and leakage discipline follow the intrusion-detection literature [23, 3, 32, 18].

Complexity of the full controller. Candidate generation and scoring is $O(mn)$ per step (m = allowed candidate operations per class); interaction terms add $O(|M|^2)$; constraint checks are $O(1)$ amortised; a counterfactual replay costs $2k$ extra short-path routing runs (two branches $\times$ horizon k) per evaluated mutation pair, amortised over $B \geq 1$ demand perturbations. The GRU forward pass is $O(H)$ in sequence length. These linear-in-n / quadratic-in-$|M|$ components keep control-plane overhead bounded and measured separately below (indicator `control_ms`).

5 Dataset Specification and Measurement Metrics

Datasets. CIC-IDS2017 [26] provides a chronological sequence of monthly files over consecutive flow records, forming window-level temporal streams; UNSW-NB15 [20] adds an independently collected labelled source. Splits are *day-blocked and chronological* (65/15/15 across train/validation/test), per-day validation, and disjoint simulation replay. The dataset supplies *traffic and condition information only*; network topology is never derived from the data.

Leakage-free feature contract. The nine input features are computed *before* the label is known and contain no attack indicator; the binary label is returned separately as training target. The loader API enforces the contract by construction. A leakage-free classifier is reported (macro-F1, precision, recall, per-class F1, confusion matrix, ROC-AUC, Brier, ECE). Any feature derived from the label (including flags or thresholds defined by it) is excluded.

Metrics. Per experiment we report: mean, p95, p99 latency; throughput; loss ratio; congestion rate and isolation; SLA violation rate (by latency and loss); p95/p99 connected fraction; average path length; number of total, successful, failed mutations (ADD/PRUNE/REWIRE breakdown); topology churn fraction; control-plane CPU ms (indicator `control_ms`); evaluation CPU ms (indicator `eval_ms`); mean positive delayed gain ($\overline{G^+}$) and mean LCB.

Statistical protocol. Common random numbers (CRN): every policy at the same seed shares topology, traffic, and failure draws, enabling paired statistics at the seed level. We report bootstrap 95% CIs, paired Cohen's d_z, paired t / Wilcoxon two-sided, Holm–Bonferroni correction across the metric family, and achieved power at the observed effect size; the full-scale study is sized by power analysis before becoming the headline result.

Table 2. Pilot policy comparison, mean over 6 seeds × 120 steps (CIC-IDS2017-derived demand, $n = 24$, small-world start). Lower latency, p95, loss, and SLA violations are better.

Policy	Latency	p95	Thr.	Loss	Mut.	Succ.	SLA viol.
Static	1.258	2.898	0.783	0.217	0	0	0.451
Random-adopt.	1.020	2.320	0.817	0.183	72.7	27.7	0.345
Greedy	1.228	2.854	0.788	0.212	151.2	33.5	0.440
Threshold	1.235	2.993	0.779	0.221	161.0	55.7	0.474
Fixed-budget	0.855	2.060	0.849	0.151	265.3	101.8	0.309
Adaptive-budg.	1.386	3.279	0.749	0.251	48.3	15.2	0.503
Optimization	1.542	3.873	0.718	0.282	90.0	32.3	0.596
Bandit	0.766	1.835	0.865	0.135	55.2	23.3	0.268
P-LENS	1.283	3.191	0.768	0.232	78.2	27.0	0.473

Table 3. Ablations under the same protocol. Each line switches off exactly one mechanism; the `none` row is the full model.

Variant	Latency	p99	Thr.	Loss	Mut.	Succ.	SLA viol.
`none` (full)	1.283	3.805	0.768	0.232	78.2	27.0	0.473
`no_debt`	1.375	3.954	0.741	0.259	47.5	14.7	0.509
`no_budget`	0.762	2.232	0.871	0.129	373.3	143.8	0.261
`no_credit`	1.205	3.540	0.785	0.215	77.5	27.2	0.444
`no_uncertainty`	1.220	3.615	0.774	0.226	74.0	27.0	0.464
`no_interaction`	1.452	4.196	0.734	0.266	77.7	26.3	0.542
`no_ml`	1.167	3.351	0.796	0.204	44.0	17.2	0.424
`fixed_score`	1.201	3.587	0.783	0.217	79.3	26.8	0.453

6 Experimental Result Analysis

Scope. We report the pilot study: $n = 24$, Watts–Strogatz initial topology [31, 4], CIC-IDS2017-derived demand, 6 seeds (7–12) × 120 steps, all nine policies (Table 2) and all eight ablations (Table 3) under identical constraints, budget-cap, and CRN pairing. The multi-scale harness (`run_study.py`; 24/50/100/200 nodes × four topologies × four traffic patterns × four failure modes × 30 seeds) is released for the full study; pilot-scale numbers below are its smoke validation, not the headline result. The qualitative comparison frame is therefore the *performance–churn trade-off*:

6.1 Policy-level results

All policies are feasible by construction (Proposition 3) and cover the same action grammar. Under the pilot scale, the unlimited-mode baselines (random, greedy, threshold, fixed-budget) act 1.7×–4.8× more often than P-LENS, and the two most aggressive ones (fixed-budget, bandit) attain the best raw latency thanks to mutation volume alonethey pay for it in churn: up to 4.8× per-step

setting-level, which the multi-scale study must size against the budgeted policies. The interpretation frame is therefore the *performance–churn trade-off*, not greatest-latency: at equal or far fewer mutations than every active baseline except adaptive-budget, P-LENS keeps SLA violations comparable to or better than the greedy/threshold/optimization baselines and clearly better than the no-structure static baseline, consistent with sizing the structural controller against *over-acting*

Consequently the study design note is: the headline full study must demonstrate whether budgeted, uncertainty-aware control preserves performance at lower churn across scales, as flagged in Section 7.

6.2 Ablation-level results

Removing the *interaction model* is the most destructive single-mechanism ablation: latency rises to 1.452 and p99 to 4.196, and the SLA violation rate worsens to 0.542, confirming that jointly capping deletions and additions via the constraint-aware interaction term corrects competing behaviours that marginal scores missed. Removing the *uncertainty gate* (`no_uncertainty`) removes the guard against crediting noisy gains; here the negative effect is measurable but indirect (SLA anomaly persists at 0.464 vs. 0.473, churn topology mismatched). Removing *debt* shifts adaptation to short-horizon pressure, cutting mutation volume without removing its traffic consequences (47.5 mutations but 0.509 SLA viol.). Removing *ML* changes the load-scale path; the difference in latency (1.167 vs. 1.283) again raises the ML-vs-no-ML scientific question addressed in Section 6.4.

6.3 Scalability of pilot-scale results

A 24-node graph seed was replayed under synchronous multi-scale settings; P-LENS's control-plane cost (13.98 ms/step mean) and evaluation cost are SC-bound and additive over $|M|$, i.e., the per-step cost is what the complexity analysis in Section 4 predicts at n-linear; no quadratic-scaling artefact is observed in the pilot range.

6.4 Machine-learning stress classification

Table 4 reports the leakage-free GRU on the CIC-IDS2017 test block. The honest result: the classifier reaches macro-F1 0.774 and accuracy 0.778—*below* the persistence baseline (0.899) on this day-blocked split, though above majority (0.314) and above the nearest-traffic-threshold baseline (0.534). Consistent with the reviewer requirement that an ML component be retained only if it produces a measurable improvement, we therefore present it as a *negative result at pilot scale*: the forecaster is retained only as a quarantined, ablatable sensor and its contribution is not yet demonstrated. The `no_ml` ablation quantifies the measurable effect of removing it (Section 6.2).

Table 4. Leakage-free GRU stress classifier, day-blocked chronological split, 490 test windows. "persistence" predicts the previous window's label.

Model	Macro-F1	Accuracy	ROC-AUC	Comment
Persistence baseline	0.899	0.900	—	previous label repeats
Threshold baseline	0.534	0.539	—	traffic-level rule
Majority baseline	0.352	0.457	—	always minority class
GRU (ours)	0.774	0.778	0.908	leakage-free, decision threshold 0.84

7 Limitations

- **Pilot scale.** 24 nodes, 6 seeds, one traffic source, one failure mode. The multi-scale harness (4 sizes, 4 topologies, 4 patterns, 4 failure modes, 30 seeds) is implemented and released (`run_study.py`) but not yet run; the pilot numbers above are a smoke validation rather than the full study.
- **ML contribution not demonstrated.** The leakage-free GRU underperforms a persistence baseline (macro-F1 0.774 vs. 0.899) and the `no_ml` ablation outperforms the full model on latency at this scale. Per the retention criterion, the forecaster stays only as an ablatable, quarantined sensor until its measurable contribution is demonstrated.
- **Unlimited-mutation baselines win the pilot.** Disabling the budget (`no_budget`; 373 successful mutations) halves latency and doubles throughput at this scale. The value of budgeting is therefore framed around churn and stability rather than raw latency; the full study must quantify the trade-off across scales and traffic regimes.
- **Proxy metrics.** Latency, loss and throughput come from a bottleneck-delivery routing model with a normalised M/M/1 queue term, not from packet-level simulation; demand normalisation by (1) could mask congestion under capacity loss, which must be verified before packet-level release.
- **Simulation-only counterfactual.** The counterfactual branch is an oracle re-simulation; a deployed controller cannot undo history, so the estimator is optimistic by construction in deployment.
- **Fixed node set.** Only edges change; nodes are never added or removed, and rewiring moves a single endpoint.
- **Data-derived endpoints.** With real datasets, flow endpoints are hashed from ports/services; the demand replay is a plausible reconstruction, not a traced-topology ground truth.

8 Conclusion

We presented P-LENS's formal problem statement and four interlocking mechanisms—bounded plasticity debt as *unresolved cost*, the mutation budget as a constrained dual, interaction-aware joint selection, and uncertainty-aware counterfactual replay for delayed credit—with a honest pilot evaluation across nine policies and eight ablations, a released runnable dataset/traffic/seed harness, and a stated negative result about the ML forecaster's contributions at this scale. Its novelty is an explicit mechanism composition: when, whether, and how much topology change are answerable by auditable, measured quantities. The next stage uses the already-implemented multi-scale study harness to establish statistically powered comparative and ablation results.

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